

Partial Derivatives and Total Differentials

ECON 441: Introduction to Mathematical Economics
Lecture 11

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Partial Differentiation

For a function of several variables:

$$y = f(x_1, x_2, \dots, x_n)$$

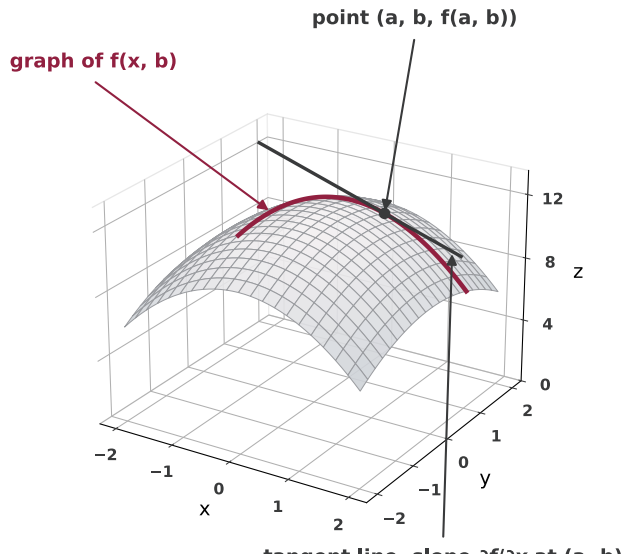
If x_1 changes by Δx_1 but all other variables remain constant:

$$\frac{\Delta y}{\Delta x_1} = \frac{f(x_1 + \Delta x_1, x_2, \dots, x_n) - f(x_1, x_2, \dots, x_n)}{\Delta x_1}$$

Partial derivative of y with respect to x_i :

$$\frac{\partial y}{\partial x_i} = f_i = \lim_{\Delta x_i \rightarrow 0} \frac{\Delta y}{\Delta x_i}$$

Partial Derivatives



Gradient Vector

Gradient: vector of all partial derivatives of a function

$$\nabla f(x_1, x_2, \dots, x_n) = [f_1, f_2, \dots, f_n]^T$$

Example

$$y = f(x_1, x_2) = 3x_1^2 + x_1x_2 + 4x_2^2$$

$$\frac{\partial y}{\partial x_1} = f_1 =$$

$$\frac{\partial y}{\partial x_2} = f_2 =$$

Example

$$y = f(u, v) = (u + 4)(3u + 2v)$$

$$\frac{\partial y}{\partial u} = f_u =$$

$$\frac{\partial y}{\partial v} = f_v =$$

Production Function

$$Q = AK^{\alpha}L^{1-\alpha}$$

Marginal product of capital (MPK):

$$\frac{\partial Q}{\partial K} = Q_K =$$

Marginal product of labor (MPL):

$$\frac{\partial Q}{\partial L} = Q_L =$$

Differentials

Note that,

$$\Delta y \equiv \left[\frac{\Delta y}{\Delta x} \right] \Delta x$$

Then for infinitesimal changes,

$$dy \equiv \left[\frac{dy}{dx} \right] dx \quad \text{or} \quad dy = f'(x)dx$$

We will call dy and dx the **differentials** of y and x , respectively.

Derivative as a Ratio

Given that,

$$dy \equiv \left[\frac{dy}{dx} \right] dx \quad \text{or} \quad dy = f'(x)dx$$

We can think of $f'(x)$ as a ratio of two quantities, dy and dx .

Elasticity

An important quantity that economists love to calculate is the **elasticity** of a function.

Elasticity is defined as:

$$\varepsilon = \frac{\text{Percentage change in } y}{\text{Percentage change in } x} = \frac{dy/y}{dx/x}$$

We can calculate this as:

$$\varepsilon = \frac{dy}{dx} \cdot \frac{x}{y}$$

Elasticity

Elasticity:

$$\varepsilon = \frac{dy}{dx} \cdot \frac{x}{y}$$

- $|\varepsilon| > 1$, elastic
- $|\varepsilon| = 1$, unit elasticity
- $|\varepsilon| < 1$, inelastic

Example

$$C = a + bY$$

Total Differential

For a function of n variables

$$y = f(x_1, x_2, \dots, x_n)$$

Total differential:

$$df = \frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2 + \dots + \frac{\partial f}{\partial x_n} dx_n = \sum_{i=1}^n f_i dx_i$$

Note: we use ∂ to tell partial derivatives apart from total derivatives. In particular,

$$\frac{\partial f}{\partial x_i} = \left. \frac{df}{dx_i} \right|_{\text{other variables are constant}}$$

Total Differential

Consider a savings function:

$$S = S(Y, i)$$

where S is savings, Y is national income, and i is the interest rate.

Total differential:

$$dS = \frac{\partial S}{\partial Y}dY + \frac{\partial S}{\partial i}di$$

Example

$$y = 5x_1^2 + 3x_2$$

Total Derivative

Total differential:

$$df = f_1 dx_1 + f_2 dx_2 + \cdots + f_n dx_n$$

We can divide the total differential by dx_1 to get the **total derivative** of f with respect to x_1 :

$$\frac{df}{dx_1} = f_1 + f_2 \cdot \frac{dx_2}{dx_1} + \cdots + f_n \cdot \frac{dx_n}{dx_1}$$

Total Derivative

Given the function

$$y = f(x_1, x_2)$$

We are interested in how y changes with respect to x_1 , but x_2 also depends on x_1 :

$$x_2 = g(x_1)$$

We know that $dy = f_1 dx_1 + f_2 dx_2$. Dividing both sides by dx_1 ,

$$\frac{dy}{dx_1} = f_1 + f_2 \cdot g'(x_1) = \frac{\partial y}{\partial x_1} + \frac{\partial y}{\partial x_2} \cdot \frac{dx_2}{dx_1}$$

A Variation on the Theme

For a function

$$y = f(x_1, x_2, w), \quad x_1 = g(w), \quad x_2 = h(w)$$

The total derivative of y is given by

$$\frac{dy}{dw} = \frac{\partial f}{\partial x_1} \frac{dx_1}{dw} + \frac{\partial f}{\partial x_2} \frac{dx_2}{dw} + \frac{\partial f}{\partial w}$$

Example

Let a production function be

$$Q(t) = A(t)K(t)^\alpha L(t)^{1-\alpha}$$

where

$$K(t) = K_0 + at \quad L(t) = L_0 + bt$$

Another Variation on the Theme

If a function is given,

$$y = f(x_1, x_2, u, v)$$

with $x_1 = g(u, v)$ and $x_2 = h(u, v)$.

Then, holding v constant, the **partial total derivative** with respect to u is

$$\frac{\partial y}{\partial u} = \frac{\partial y}{\partial x_1} \frac{\partial x_1}{\partial u} + \frac{\partial y}{\partial x_2} \frac{\partial x_2}{\partial u} + \frac{\partial y}{\partial u}$$

Homework Questions

- Exercise 7.4: 1 (a), (d), 2 (a), (b), 3, 5, 7
- Exercise 8.1: 1 (a), 4, 6
- Exercise 8.2: 3 (a), 4, 5, 6, 7 (b), (f)
- Exercise 8.4: 2, 4

Textbook reference: 7.4, 8.1, 8.2, 8.4

Reminder: Quiz 2 on Monday.